Identification of Cattle's Breed Type by Processing Image using Deep Learning Models

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2024 International Conference on Integrated Intelligence and Communication Systems (ICIICS) | 979-8-3315-0496-0/24/$31.00 ©2024 IEEE | DOI: 10.1109/ICIICS63763.2024.10859520

***Abstract*—Accurately identifying cattle breeds is crucial for effective livestock management and breeding. This research work uses advanced computer techniques to automatically recognize cattle breeds from images. It was used with Convolutional Neural Networks (CNNs), which are good at detecting specific details in images to distinguish one breed from another. Before predicting, images are prepared and enhanced to ensure accurate results. The research work tests CNN models like DenseNet201, MobileNetV2, and Xception to determine which performs bettter. It measured performance using metrics like precision, recall, and F1 score. This automated approach provides farmers and breeders with a reliable tool to identify breeds and improve management. CNN technology helps modernize traditional farming methods, supporting sustainable livestock care. The study aims to find the most accurate CNN model for predicting cattle breeds, emphasizing the benefits of automated systems in maintaining breed purity, genetic diversity, and overall livestock health.**

Keywords—cattle breed, CNN, densenet201, mobilenetv2, xception, inceptionnet v3, precision, f1 score.

I. Introduction

Cattle breed identification plays a major role in modern livestock management and breeding practices. Accurately discerning between different cattle breeds according to their visual characteristics is essential for maintaining breed purity and optimizing breeding strategies. Traditionally, manual inspection is often used, which can be slow and susceptible to human mistakes. To address this, this article initiates deep learning, particularly CNN, to streamline the process of identifying cattle breeds through image analysis. CNNs are well-known for their ability to extract detailed features from images, which makes them ideal for discerning subtle differences in cattle breeds based on visual cues such as coat color, horn shape, and body proportions. This work explores the efficiency of several CNN architectures, including DenseNet201, MobileNetV2, and Xception, in accurately predicting and classifying cattle breeds from image datasets. Each architecture is assessed using metrics like precision, recall and F1 score to identify the model that attains the highest accuracy in breed identification. The primary advantage of employing deep learning in this context is its capacity to provide a scalable and automated solution to cattle breed identification. By automating this process, farmers and breeders can simplify their processes, cut down on manual work, and enable more informed choices about breeding and herd management. Moreover, accurate breed identification enhances genetic diversity preservation efforts and supports sustainable livestock agricultural methods. This research work is designed to demonstrate how innovative technologies such as deep learning have the ability to revolutionize conventional agricultural practices leading to more efficient and effective management of cattle herds. It aims to develop an automated system for predicting cow breeds based on image processing and machine learning techniques. The process begins with the acquisition of cow images, followed by data preprocessing steps such as noise removal, gray scale conversion, and image enhancement to ensure high-quality inputs for analysis. Key features are extracted from the processed images, which are then used to train and test a breed classification model. The ultimate goal is to achieve a breed prediction system that can assist farmers and breeders in managing livestock more effectively by helping advanced image processing and artificial intelligence technologies.

II. Literature Survey

The literature review investigates innovative solutions for cattle breed identification, gaining insights from different research studies that explore image-based identification techniques, feature extraction methods, and advancements in deep learning models for breed classification. The study [1] examines traditional methods such as RFID and visual tagging, highlighting their invasive nature and the necessity for physical handling of animals. It advocates for non-invasive image-based identification techniques as a superior alternative for cattle breed identification. The work [2] delves into the application of image datasets for cattle breed identification, focusing on pre-processing steps like dimension standardization and noise removal. It highlights the application of the Scale-Invariant Feature Transform (SIFT) method for extracting essential features from cattle images enhancing classification accuracy. The study [3] investigates how CNN can be used to classify cattle breeds into multiple categories. It underscores the high accuracy of CNNs in image recognition tasks and their potential in handling complex breed identification scenarios. The article [4] explores the integration combining machine learning approaches with conventional image processing techniques, demonstrating improved accuracy and robustness in cattle breed classification. The work [5] investigates the application of transfer learning in cattle breed identification, utilizing pre-trained models on large datasets. It significantly reduces training time and improves classification accuracy by leveraging existing knowledge from related tasks. Research work [6] emphasizes the significance of having high-quality and diverse datasets when training deep learning models for breed identification. It demonstrates that large instead of ear tagging or RFID, it employs CNNs to classify breeds from various angles, improving accuracy over traditional methods. The work [7] introduces a multi-view approach to cattle breed identification, using images from different angles to capture comprehensive aspects of the cattle. It improves identification accuracy by providing a more holistic view of the animal. The article [8] evaluates the performance of different CNN architectures for cattle breed identification. It compares the effectiveness of different models and determines the most appropriate architecture for this particular task. A study [9] concentrates on applying data augmentation methods to

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improve the effectiveness of CNNs in cattle breed identification. By artificially increasing the size and variety in the training dataset, the study achieves better model generalization and accuracy. The work [10] examines the application of real-time image analysis and classification systems for cattle breed identification. It highlights the capability of deploying these systems in practical settings to enable immediate and accurate breed identification within the domain.The work [11] focuses on using ResNet architectures for cattle breed identification, highlighting their deeper network structures and improved performance over traditional CNNs. The study [12][17] investigates the use of histogram of oriented gradients (HOG) features in combination with deep learning models, demonstrating enhanced breed classification accuracy. The work [13][18] explores the impact of different image resolutions on the performance of breed identification models, finding that higher resolution images significantly improve accuracy. The article [14][19] presents a novel method of using generative adversarial networks (GANs) to create synthetic cattle images, which helps in augmenting training datasets and improving model robustness. Research [15][20] delves into the use of attention mechanisms in CNNs to focus on relevant features of cattle images, thereby improving classification accuracy. The study [16][21][22] explores the combination of traditional feature extraction methods like SIFT and SURF with deep learning models, showing improved performance in breed identification.

In particular, these studies converge on modernizing cattle breed identification by leveraging non-invasive image-based techniques, advanced feature extraction methods, along with deep learning models. They collectively endorse dynamic and robust systems that address traditional limitations, streamline the identification process, and improve the overall accuracy and efficiency of cattle breed classification.

III. Methodology

The proposed system aims to identify cattle breeds using deep learning techniques. The dataset consists of 1208 images sourced from Kaggle, representing five cattle breeds: Ayrshire, Brown Swiss, Holstein Friesian, Jersey, and Red Dane. These images have been captured using mobile phones and cameras. Capturing various views and instances where multiple cows may appear in a single image.

A. Dataset

The dataset sourced from Kaggle contains 1208 images representing five cattle breeds. This model utilized a dataset of over 1208 images, which was categorized into five cattle types from different regions. Images similar to those shown in Fig.1 were included in the dataset. The model identifies five different cattle breeds: Ayrshire, Brown Swiss, Holstein Friesian, Jersey and Red Dane

Fig. 1. Dataset for cattle breed identification

The methodology applied in this research is illustrated in the flowchart provided in Fig 2.

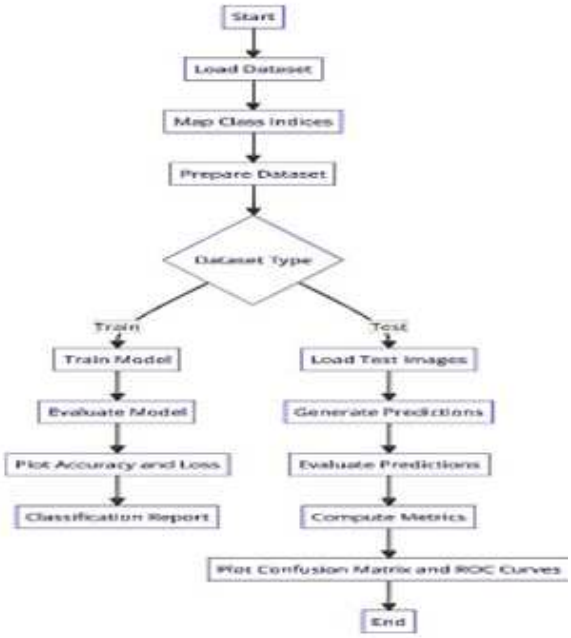

Fig. 2. Flowchart outlining the methodology steps

The flow chart shown in Fig. 2 is a CNN designed platform for cattle breed identification begins with convolutional layers to extract aspects such as edges and textures from the input image. Pooling layers reduce dimensionality while preserving key information. Further convolutional and pooling layers refine the features. A softmax layer classifies the image into the most likely breed category, producing the final output. The Shows Diagram of the proposed system architecture in fig. 3.

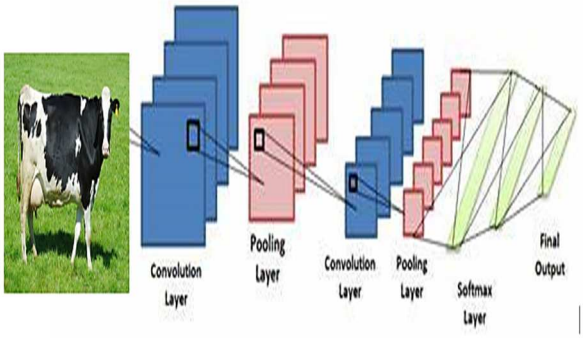

Fig. 3. Diagram of the proposed system architecture

Steps:

- Data Collection and preparation Data Preprocessing
- Data Augmentation
- Model Building
- Training and Evaluation

1) Data Collection and preparation: The dataset, consisting of 1208 images representing five different cattle breeds (Ayrshire, Brown Swiss, Holstein Friesian, Jersey, and Red Dane), was obtained from Kaggle. The images are split into training and testing groups. Each image is resized to 180x180 pixels, converted into arrays, and normalized to enhance the model's effectiveness. The labels are encoded

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into a structure suitable for multi-class classification. The dataset is subsequently divided into training and testing subsets to evaluate the accuracy of the model.

2) Data Preprocessing: Before inputting the data into the model, various preprocessing procedures are performed to improve image quality and facilitate accurate classification. These steps include removing noise, converting images to gray scale, and other enhancements. This stage guarantees that the images are cleaned and standardized, facilitating the extraction of important features by deep learning models. The Shows Input image with noise and background removed in fig. 4.

Fig. 4. Input image with noise and background removed

3) Data Augmentation: To enhance the model's robustness and performance, data augmentation methods are used, such as horizontal and vertical flips, rotations, zooming, width and height shifts, and shear transformations. This process generates more diverse training data, aiding the model in better generalizing to new, unseen images.

4) Model building: For breed classification, models like DenseNet201, MobileNet V2, Inception V3, and Xception are used. DenseNet201 enhances feature reuse, while MobileNet V2 is efficient for mobile applications. Inception V3 captures multi-scale features, and Xception employs depth wise separable convolutions for improved performance. These models are trained using augmented data, using dropout and batch normalization to prevent over fitting.

a) *DenseNet 201:* It works by connecting each layer in the network to every other layer, which allows the model to reuse features and share information more effectively. Instead of just passing data from one layer to the next, DenseNet combines the outputs from all previous layers and uses them as inputs for each new layer. This design helps the model learn more complex patterns, improves the flow of information, and makes training more efficient. As a result, DenseNet can achieve high accuracy while using fewer parameters compared to other models, making it both powerful and efficient for tasks like image classification.

The Fig. 5 depicts the confusion matrix which shows how the model classifies different cattle breeds. Correct predictions are shown along the diagonal, like 84 Ayrshire and 95 Holstein Friesian cattle identified correctly. Misclassifications, such as 15 Ayrshire cattle being labeled as Brown Swiss, are shown off-diagonal. The color intensity reflects the number of predictions, with yellow indicating higher accuracy. This matrix highlights the model's strengths and areas for improvement in breed identification.

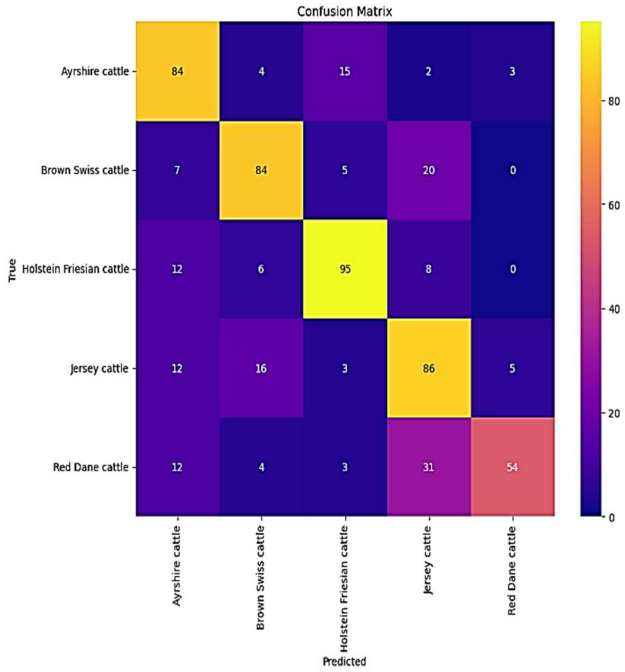

Fig. 5. Confusion Matrix for multi-class classification using DenseNet201

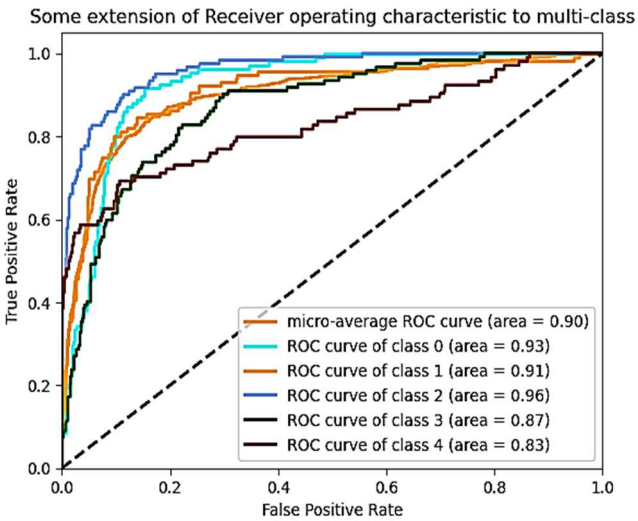

Fig. 6. ROC Curves for Multi-class Classification Using DenseNet201

The image shown in the Fig. 6 depicts a multi-class ROC curve, representing the model's performance across different classes. Each line corresponds to a class, with AUC values indicating accuracy—class 2 has the highest AUC (0.96) and class 4 the lowest (0.83). The overall performance is summarized by the micro-average ROC curve with an AUC of 0.90. The curves above the diagonal indicate effective classification.

The image shown in Fig. 7 depicts loss and accuracy graphs by comparing training and validation metrics over 30 epochs. The left graph displays accuracy, with both training and validation accuracy improving over time. The right graph shows loss, with both losses decreasing, though validation loss

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remains higher, indicating slight overfitting. These graphs reflect the model's learning progress.

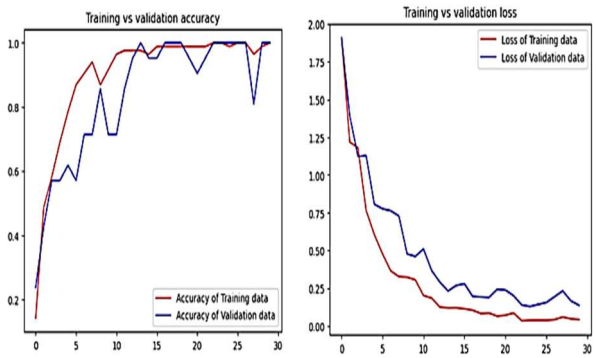

Fig. 7. Loss and Accuracy graph for DenseNet201

b) Inception V3: It is a deep learning model for image classification that uses Inception modules, which apply multiple filters simultaneously to capture features at different scales. It also uses factorized convolutions to improve efficiency while maintaining accuracy, making it effective for complex image recognition tasks.

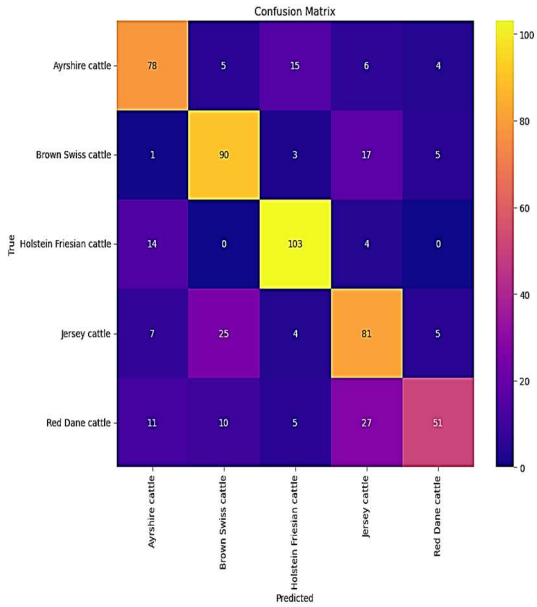

Fig. 8. Confusion Matrix for Cattle Breed Classification Using InceptionV3

The confusion matrix shown in Fig. 8 depicts the model's performance across five cattle breeds. The diagonal represents correct predictions, while off-diagonal cells show misclassifications. For instance, Holstein Friesian cattle had 103 correct predictions, and Ayrshire cattle had 15 misclassified as Holstein Friesian. This matrix highlights the model's accuracy and areas of misclassification.

The ROC curve shown in Fig. 9 illustrates the performance of a multi-class classifier across five different classes. Each curve represents a class's true positive rate against its false positive rate, with the area under the curve (AUC) indicating the classifier's accuracy for that class. The micro-average ROC curve, which combines all classes, has an AUC of 0.90, showing good overall performance. Individual class AUCs range from 0.86 to 0.95, indicating varying levels of classification accuracy across the different classes.

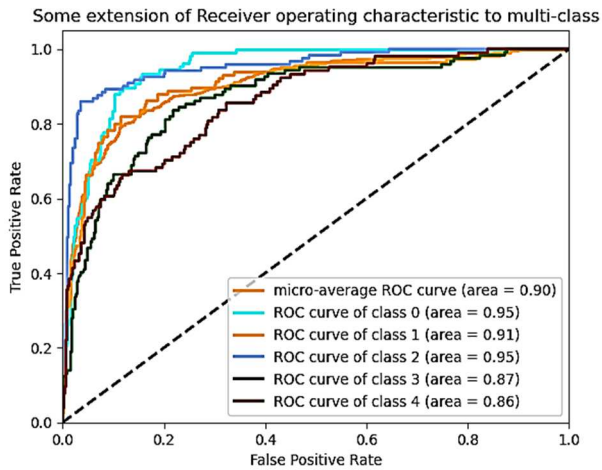

Fig. 9. ROC Curves for Multi-class Classification Using InceptionV3

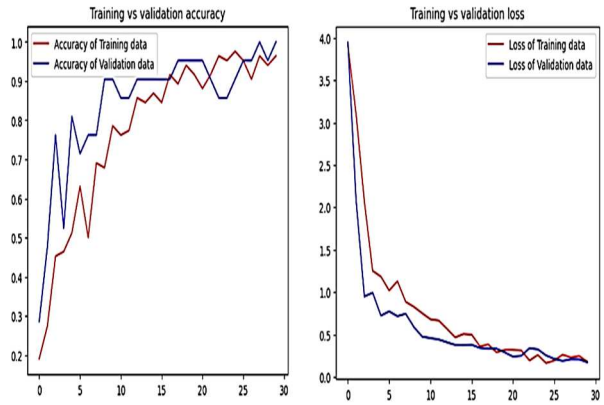

Fig. 10. Graph of Loss and Accuracy for Inception V3

In the Fig. 10, the left plot shows the accuracy of training and validation data over 30 epochs, where both accuracies increase steadily, with training accuracy approaching 100%. Initially, the validation accuracy fluctuates but later stabilizes at a high value, indicating good generalization. The right plot shows the training and validation loss, both decreasing sharply in the early epochs, and eventually converging to a low, stable value, suggesting the model is well-fitted to the data.

c) MobileNet V2: It is a deep learning model designed to work efficiently on mobile devices and small systems. The model uses fewer calculations to learn complex features, making it ideal for situations where both speed and accuracy are important, especially when there are limited resources.

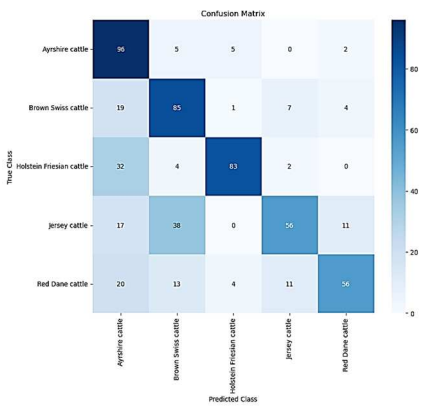

Fig. 11. Confusion Matrix for Cattle Breed Classification Using MobileNet V2

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The Fig. 11 shows the confusion matrix which depicts the model's performance in classifying five cattle breeds. Ayrshire cattle were mostly accurate with 96 correct predictions, while Brown Swiss had 85 correct but 19 misclassified as Ayrshire. Holstein Friesian had 83 correct predictions, though 32 were wrongly classified as Ayrshire. Jersey cattle were harder to classify, with only 56 correct and 38 misclassified as Brown Swiss. Red Dane cattle had 56 correct predictions, but 20 were mistaken for Ayrshire, highlighting areas for improvement.

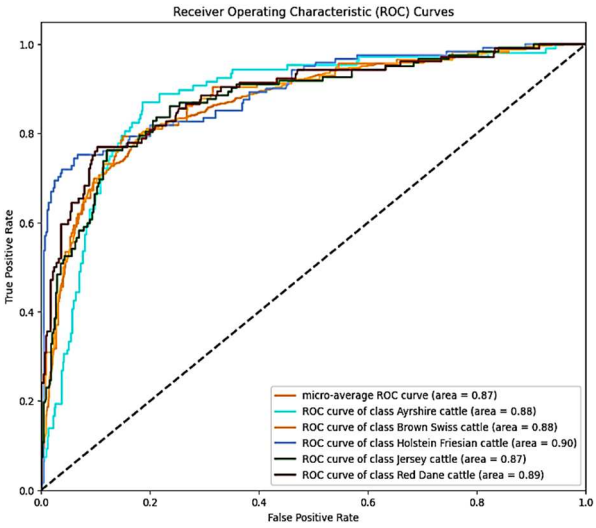

Fig. 12. ROC Curves for Multi-class Classification Using MobileNet V2

The Fig. 12 shows ROC curves which depicts the performance of the cattle breed classification model across six classes, including Ayrshire, Brown Swiss, Holstein Friesian, Jersey, and Red Dane cattle. The AUC values indicate how well the model distinguishes between breeds, with Holstein Friesian achieving the highest AUC of 0.90, followed by Red Dane (0.89) and Ayrshire/Brown Swiss (both 0.88). The overall micro-average AUC is 0.87, reflecting strong classification performance.

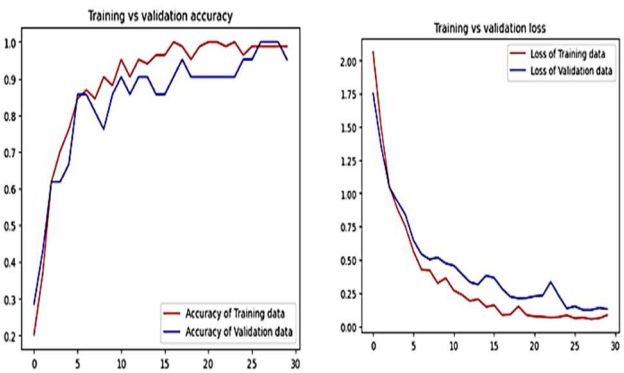

Fig. 13. Graph of Loss and Accuracy for MobileNet V2

The Fig. 13 shows the graphs which compares the training and validation accuracy (left) and loss (right) over 30 epochs. The accuracy plot shows that both training and validation accuracy increase over time, stabilizing near 1.0, indicating good model performance. The loss plot shows that both training and validation loss decrease sharply in the initial epochs and gradually level off, indicating effective learning with minimal overfitting.

d) Xception: It is a deep learning model that improves on Inception by using a special type of convolution called depth wise separable convolution. This method splits the process into two parts: one for focusing on spatial details and another for channel-wise operations. This design helps the model learn more efficiently, making it great for image classification with less computational effort.

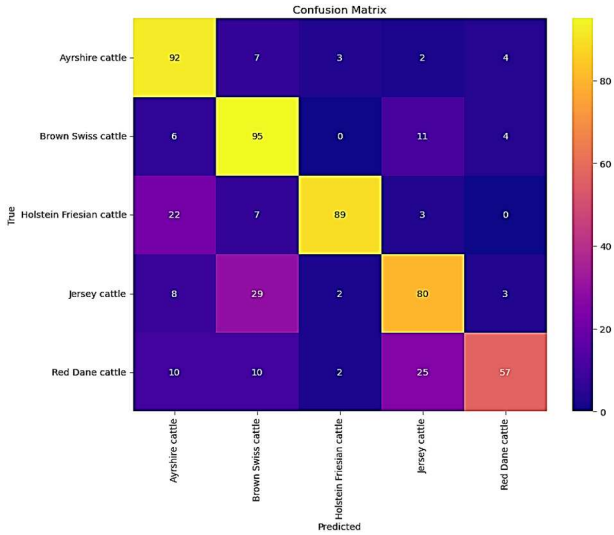

Fig. 14. Confusion Matrix for Cattle Breed Classification Using Xception

The Fig. 14 shows the confusion matrix which depicts the performance of a model in classifying five cattle breeds: Ayrshire, Brown Swiss, Holstein Friesian, Jersey, and Red Dane. The diagonal cells represent correct classifications, while off-diagonal cells show misclassifications. For example, the model correctly classified 92 Ayrshire cattle but misclassified some as other breeds. Each row totals the true instances of a breed, while each column totals the predicted instances. The color intensity indicates the number of cases, with brighter colors representing higher values.

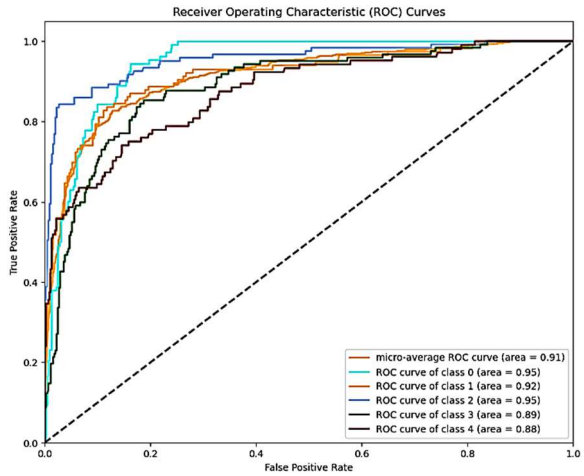

Fig. 15. ROC Curves for Multi-class Classification using Xception

The Fig. 15 shows the ROC curve which depicts the model's performance in classifying five cattle breeds. Each curve represents the balance between true positive and false positive rates for a class, with AUC values indicating effectiveness. Higher AUC values mean better performance. The micro-average ROC curve summarizes the model's overall accuracy (AUC = 0.91).

The image shown in Fig. 16 depicts loss and accuracy graphs for xception. The left graph compares training and

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validation accuracy, both increasing over 30 epochs with similar trends. The right graph compares training and validation loss, both decreasing steadily, indicating effective learning and minimal overfitting.

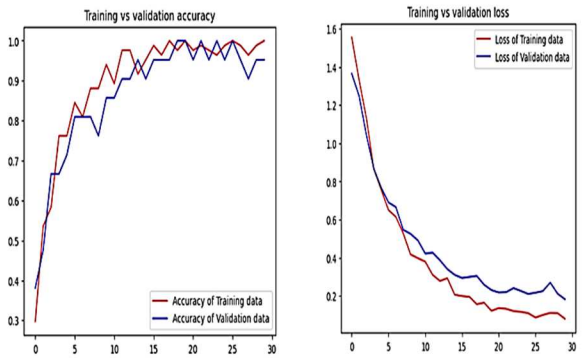

Fig. 16. Graph of Loss and Accuracy for Xception

5) Training and Evaluation: In this research work, the dataset consists of 1,208 images of five cattle breeds. The dataset is divided into three sets: training, validation, and test sets. Typically, 70% of the data is used for training, 15% for validation, and 15% for testing. This split ensures that the model has enough data to learn from, while also keeping sufficient data to validate and test the model's performance on unseen images.The models are trained with the augmented training data using a suitable optimizer and loss function. After the training process is finished, the models are assessed on the testing set to measure their accuracy, precision, recall, F1 score, and ROC-AUC. The evaluation ensures that the models generalize well to new, unseen images. Additionally, confusion matrices and ROC curves are plotted to examine the performance of each model, highlighting areas for potential improvement.

IV. Results and Discussions

TABLE I. Performance Metrics Of Deep Learning Models For Cattle Breed Identification

Models	Accuracy in %	Precision in %	Recall in %	F1 Score in %
DenseNet121	70.4	73.9	69.8	69.7
MobileNet V2	65.6	66.8	65.4	65.5
Xception	72.3	74.8	72.1	72.0
InceptionNet	70.2	72.6	69.7	69.5

The Table 1 shows the performance metric of deep learning models for identification of cattle breeds. It evidenced from the table that accuracy obtained for the models DenseNet121, MobileNetV2, Xception and InceptionNet are 70.4%, 65.6%, 72.3% and 70.2% respectively. Similarly, precisions are 73.9%, 66.8%,74.8 and 72.6% respectively. The recall factors are 69.8%, 65.4%, 72.1 and 69.7 respectively. The F1 scores are 69.7%, 65.5%, 72.0 and 69.5% respectively. As for as performance is considered, the Xception model has got good result compared to rest of the models. The second good one is DenseNet121 followed by InceptionNet and MobileNetV2.

The accuracy model shown in the Fig. 17 illustrates the performance of DenseNet201, MobileNet V2, Xception and Inception was evaluated using accuracy, precision, recall, F1 score, and ROC-AUC. It was evidenced that DenseNet201 achieved the highest accuracy and F1 score due to its dense connections. MobileNet V2, while efficient, had slightly lower recall. Inception excelled in multi-scale feature capture, and Xception benefited from depth wise separable convolutions. Misclassifications often occurred between similar breeds like Ayrshire and Brown Swiss. All models had high AUCs. Further preprocessing, augmentation, and model fine-tuning could improve breed-specific accuracy. DenseNet201 shows significant promise for cattle breed identification. Also, when comparing models, it should be consider more than accuracy. For example, while DenseNet201 may be more accurate, it could be slower to train, and MobileNetV2 might be faster and better for mobile use but with slightly lower accuracy. By looking at the pros and cons of each model in terms of speed, resource use, and performance, it can give a better idea of which one is most useful.

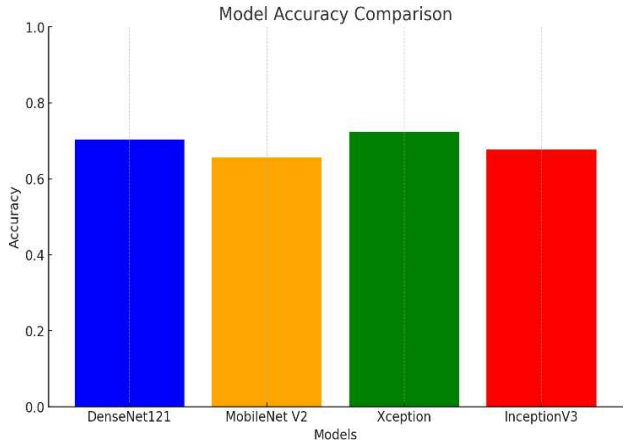

Fig. 17. Accuracy of Various Deep Learning Models for Cattle Breed Identification

V. Conclusion

This research work aims to make identifying cattle breeds easier and more accurate by using one of the advanced computer techniques called CNN. It automatically analyze the images to recognizedifferent cattle breeds. Using a dataset from Kaggle with 1208 images of five breed types Ayrshire, Brown Swiss, Holstein Friesian, Jersey and Red Dane. This system tests different CNN models like DenseNet201, MobileNetV2, InceptionV3, and Xception. The images are processed to improve quality and ensure consistency by resizing, normalizing, and augmenting. The models have been trained using techniques to enhance performance and prevent errors, including techniques like data augmentation, dropout, and batch normalization. They are then evaluated using precision, recall, and F1 scoreto measure their accuracy. This approach helps farmers and breeders by saving time, reducing manual work, and ensuring better cattle management. It supports sustainable farming by preserving genetic diversity and maintaining breed purity, showing how modern technology can improve traditional farming methods.

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